Session 5 Homework Solutions

APSTA-GE 2006: Applied Statistics for Social Science Research

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Table of contents

This document provides worked solutions to the Lecture 5 homework.

1 Binomial PMF and CDF

> **Task:** Let $X \sim \text{Binomial}(20, 0.35)$. Compute $\mathbb{P}(X = 7)$, $\mathbb{P}(X \leq 7)$, and $\mathbb{P}(5 \leq X \leq 10)$; verify each result with `dbinom()` or `pbinom()`, then plot the PMF and CDF for $x = 0, \ldots, 20$.

For a binomial random variable with $N = 20$ and $\theta = 0.35$, the PMF is

$$\mathbb{P}(X = x) = \binom{20}{x}(0.35)^x(0.65)^{20-x}, \qquad x = 0, 1, \ldots, 20.$$

Therefore,

$$\mathbb{P}(X = 7) = \binom{20}{7}(0.35)^7(0.65)^{13}.$$

The two interval probabilities can be written as

$$\begin{aligned}\mathbb{P}(X \leq 7) &= \sum_{x=0}^{7}\binom{20}{x}(0.35)^x(0.65)^{20-x},\\ \mathbb{P}(5 \leq X \leq 10) &= \mathbb{P}(X \leq 10) - \mathbb{P}(X \leq 4).\end{aligned}$$

```r
N <- 20
theta <- 0.35

p_equal_7 <- dbinom(7, size = N, prob = theta)
p_at_most_7 <- pbinom(7, size = N, prob = theta)
p_5_through_10 <-
  pbinom(10, size = N, prob = theta) -
  pbinom(4, size = N, prob = theta)

binomial_answers <- data.frame(
  quantity = c("P(X = 7)", "P(X <= 7)", "P(5 <= X <= 10)"),
  probability = c(p_equal_7, p_at_most_7, p_5_through_10)
)

knitr::kable(
  binomial_answers,
  digits = 5,
  booktabs = TRUE,
  caption = "Binomial probability calculations"
)
```

Table 1: Binomial probability calculations

quantity	probability
P(X = 7)	0.18440
P(X <= 7)	0.60103
P(5 <= X <= 10)	0.82864

Thus,

$$\boxed{\mathbb{P}(X = 7) = 0.18440, \quad \mathbb{P}(X \leq 7) = 0.60103, \quad \mathbb{P}(5 \leq X \leq 10) = 0.82864}.$$

The PMF gives each point probability, while the CDF accumulates those probabilities from left to right.

```r
binomial_grid <- data.frame(
  x = 0:N,
  pmf = dbinom(0:N, size = N, prob = theta),
  cdf = pbinom(0:N, size = N, prob = theta)
)

pmf_plot <- ggplot(binomial_grid, aes(x, pmf)) +
  geom_col(width = 0.7, fill = "#3564d4") +
  scale_x_continuous(breaks = seq(0, N, by = 2)) +
  labs(
    x = "x",
    y = "P(X = x)",
    title = "Probability mass function"
```

```
  )

cdf_plot <- ggplot(binomial_grid, aes(x, cdf)) +
  geom_step(direction = "hv", linewidth = 0.6, color = "#d95f02") +
  geom_point(size = 1.3, color = "#d95f02") +
  scale_x_continuous(breaks = seq(0, N, by = 2)) +
  scale_y_continuous(limits = c(0, 1)) +
  labs(
    x = "x",
    y = "P(X <= x)",
    title = "Cumulative distribution function"
  )

pmf_plot + cdf_plot
```

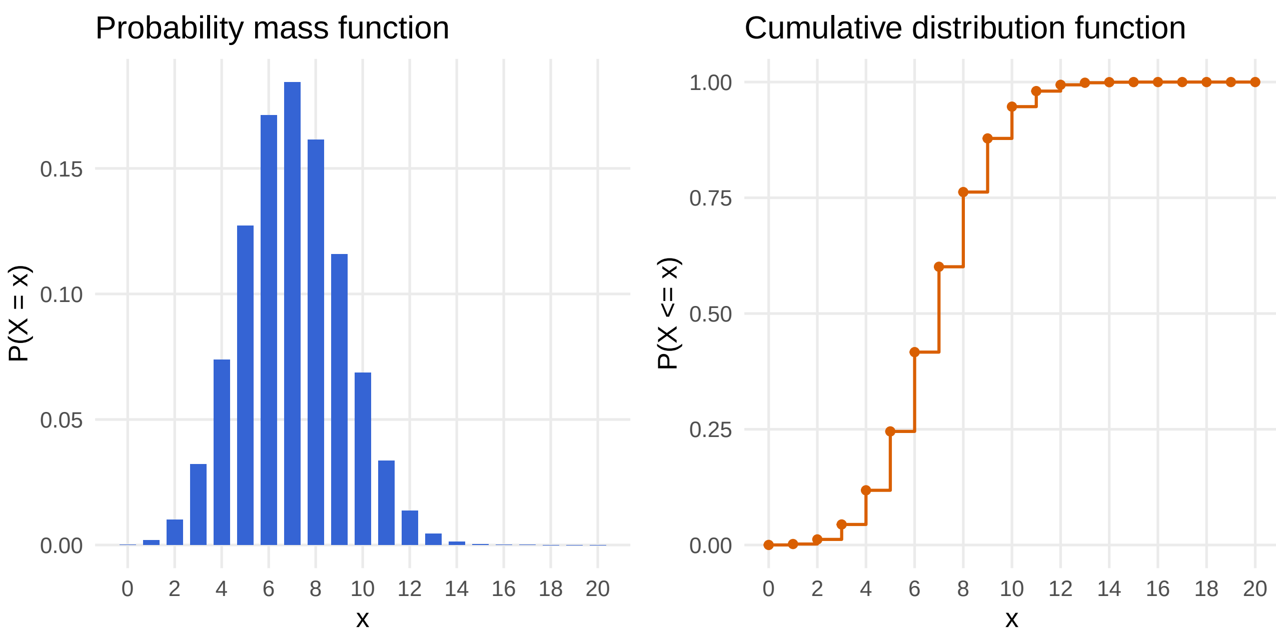

Figure 1: PMF and CDF of a Binomial(20, 0.35) random variable.

2 First-success simulation

> **Task:** Let T be the trial on which the first success occurs when $\theta = 0.2$. Simulate 50,000 values with `rgeom() + 1`; for $t = 1, \dots, 10$, compare the empirical probabilities with $\mathbb{P}(T = t) = \theta(1-\theta)^{t-1}$. Compare the simulated mean and variance with $1/\theta$ and $(1-\theta)/\theta^2$.

R's `rgeom()` counts the number of failures before the first success. Adding 1 changes that output into the trial number of the first success.

For $T = 1, 2, \dots,$

$$\mathbb{P}(T = t) = \theta(1-\theta)^{t-1}, \qquad \mathbb{E}(T) = \frac{1}{\theta}, \qquad \mathbb{V}(T) = \frac{1-\theta}{\theta^2}.$$

With $\theta = 0.2$, the theoretical mean and variance are

$$\mathbb{E}(T) = 5, \qquad \mathbb{V}(T) = 20.$$

```r
set.seed(502)

geometric_theta <- 0.2
geometric_reps <- 50000
first_success <- rgeom(geometric_reps, prob = geometric_theta) + 1

estimate_empirical_probability <- function(value, simulated_values) {
  number_of_matches <- sum(simulated_values == value)
  number_of_simulations <- length(simulated_values)

  number_of_matches / number_of_simulations
}

geometric_comparison <- data.frame(t = 1:10) |>
  mutate(
    empirical = purrr::map_dbl(
      t,
      estimate_empirical_probability,
      simulated_values = first_success
    ),
    theoretical = geometric_theta * (1 - geometric_theta)^(t - 1),
    difference = empirical - theoretical
  )

knitr::kable(
  geometric_comparison,
  digits = 5,
  booktabs = TRUE,
  caption = "Empirical and theoretical first-success probabilities"
)
```

Table 2: Empirical and theoretical first-success probabilities

t	empirical	theoretical	difference
1	0.19802	0.20000	-0.00198
2	0.15734	0.16000	-0.00266
3	0.12846	0.12800	0.00046
4	0.10434	0.10240	0.00194
5	0.08062	0.08192	-0.00130
6	0.06732	0.06554	0.00178
7	0.05222	0.05243	-0.00021

t	empirical	theoretical	difference
8	0.04186	0.04194	-0.00008
9	0.03288	0.03355	-0.00067
10	0.02668	0.02684	-0.00016

The pointwise simulation error is small and changes sign across support points, as expected from Monte Carlo variation.

```r
geometric_comparison |>
  select(t, empirical, theoretical) |>
  pivot_longer(
    cols = c(empirical, theoretical),
    names_to = "source",
    values_to = "probability"
  ) |>
  ggplot(aes(t, probability, color = source, shape = source)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 1:10) +
  scale_color_manual(
    values = c(empirical = "#3564d4", theoretical = "#d95f02"),
    labels = c(empirical = "simulation", theoretical = "theory")
  ) +
  scale_shape_manual(
    values = c(empirical = 16, theoretical = 1),
    labels = c(empirical = "simulation", theoretical = "theory")
  ) +
  labs(
    x = "trial of first success, t",
    y = "probability",
    color = NULL,
    shape = NULL
  )
```

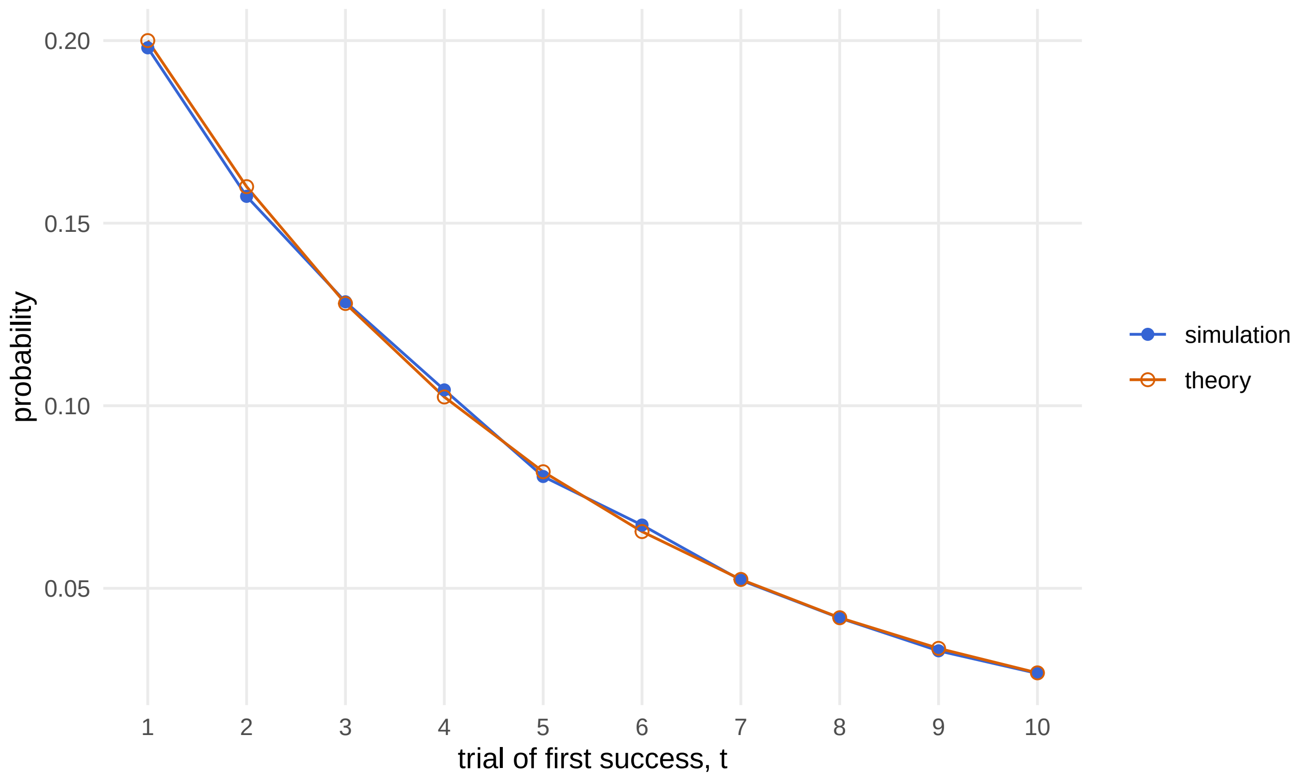

Figure 2: Simulated and theoretical probabilities for the first-success trial.

```
geometric_moments <- data.frame(
  source = c("simulation", "theory"),
  mean = c(mean(first_success), 1 / geometric_theta),
  variance = c(
    var(first_success),
    (1 - geometric_theta) / geometric_theta^2
  )
)

knitr::kable(
  geometric_moments,
  digits = 4,
  booktabs = TRUE,
  caption = "First-success mean and variance"
)
```

Table 3: First-success mean and variance

source	mean	variance
simulation	5.0369	20.1046
theory	5.0000	20.0000

The simulated mean is 5.037 and the simulated variance is 20.105, both close to 5 and 20.

3 Rare events: binomial and Poisson

> **Task:** Simulate 100,000 values from Binomial$(1000, 0.003)$ and compare the empirical PMF for $0, \dots, 10$ with a Poisson PMF having $\lambda = 3$. Compare the two means, variances, and probabilities of at least 6 events.

Here

$$N = 1000, \qquad \theta = 0.003, \qquad N\theta = 3.$$

The binomial mean and variance are

$$\mathbb{E}(X) = N\theta = 3, \qquad \mathbb{V}(X) = N\theta(1-\theta) = 2.991.$$

A Poisson(3) random variable has mean and variance equal to 3, so the two distributions should be close.

```
set.seed(503)

rare_reps <- 100000
rare_size <- 1000
rare_prob <- 0.003
poisson_lambda <- rare_size * rare_prob

rare_counts <- rbinom(
  rare_reps,
  size = rare_size,
  prob = rare_prob
)

# Reuse the empirical-probability function from Problem 2
rare_pmf <- data.frame(k = 0:10) |>
  mutate(
    empirical_binomial = purrr::map_dbl(
      k,
      estimate_empirical_probability,
      simulated_values = rare_counts
    ),
    exact_binomial = dbinom(k, size = rare_size, prob = rare_prob),
    poisson = dpois(k, lambda = poisson_lambda)
  )

knitr::kable(
  rare_pmf,
  digits = 5,
  booktabs = TRUE,
  caption = "Rare-event PMF comparison"
)
```

Table 4: Rare-event PMF comparison

k	empirical_binomial	exact_binomial	poisson
0	0.04978	0.04956	0.04979
1	0.14872	0.14914	0.14936
2	0.22280	0.22415	0.22404
3	0.22404	0.22438	0.22404
4	0.16857	0.16828	0.16803
5	0.10172	0.10087	0.10082
6	0.05027	0.05033	0.05041
7	0.02216	0.02151	0.02160
8	0.00835	0.00803	0.00810
9	0.00254	0.00266	0.00270
10	0.00083	0.00079	0.00081

```r
rare_pmf |>
  pivot_longer(
    cols = -k,
    names_to = "source",
    values_to = "probability"
  ) |>
  ggplot(aes(k, probability, color = source, shape = source)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.8) +
  scale_x_continuous(breaks = 0:10) +
  scale_color_manual(
    values = c(
      empirical_binomial = "#222222",
      exact_binomial = "#3564d4",
      poisson = "#d95f02"
    ),
    labels = c(
      empirical_binomial = "simulated binomial",
      exact_binomial = "exact binomial",
      poisson = "Poisson comparison"
    )
  ) +
  scale_shape_manual(
    values = c(
      empirical_binomial = 16,
      exact_binomial = 1,
      poisson = 2
    ),
    labels = c(
      empirical_binomial = "simulated binomial",
      exact_binomial = "exact binomial",
      poisson = "Poisson comparison"
    )
  ) +
```

```
  labs(
    x = "event count",
    y = "probability",
    color = NULL,
    shape = NULL
  )
```

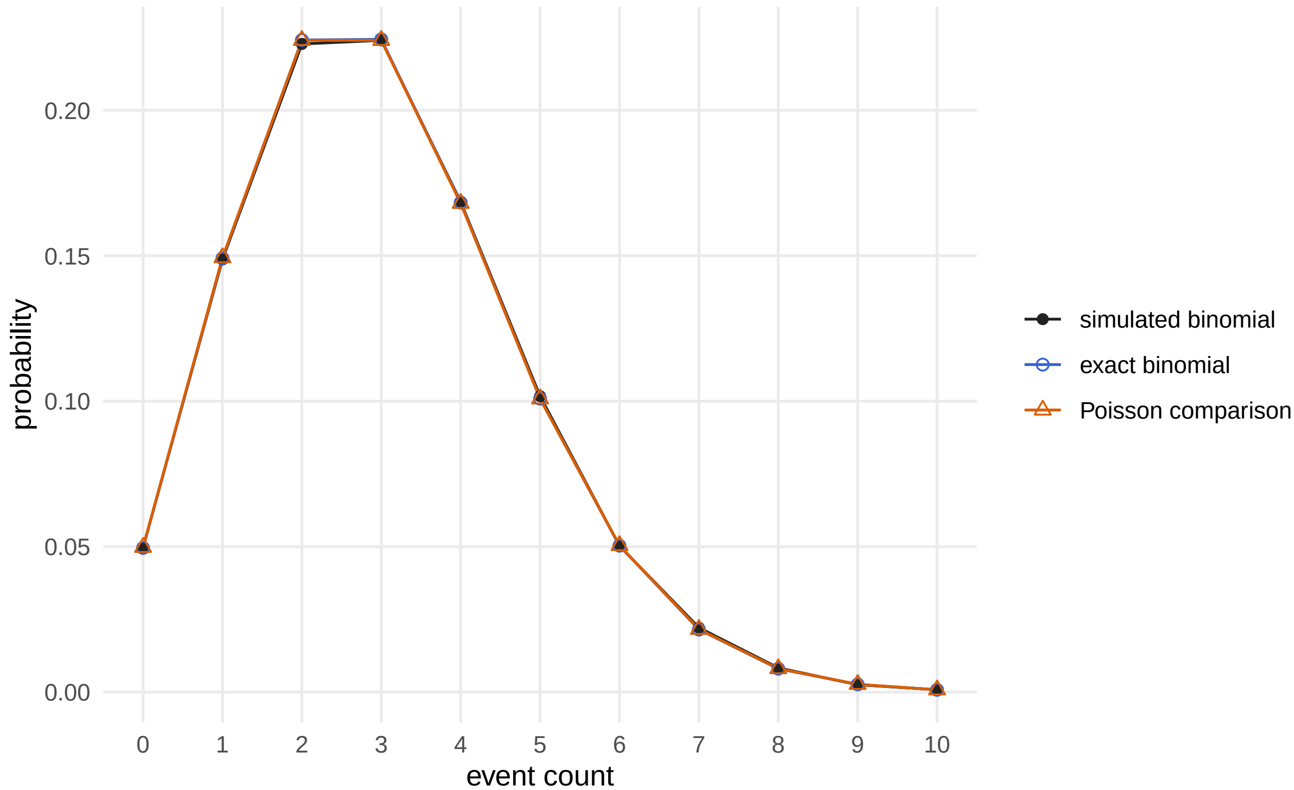

Figure 3: The empirical binomial PMF and its Poisson comparison.

For the upper-tail probability,

$$\mathbb{P}(X \geq 6) = 1 - \mathbb{P}(X \leq 5).$$

```
rare_summary <- data.frame(
  source = c(
    "simulated binomial",
    "exact binomial",
    "Poisson comparison"
  ),
  mean = c(
    mean(rare_counts),
    rare_size * rare_prob,
    poisson_lambda
  ),
```

```
  variance = c(
    var(rare_counts),
    rare_size * rare_prob * (1 - rare_prob),
    poisson_lambda
  ),
  probability_at_least_6 = c(
    mean(rare_counts >= 6),
    pbinom(5, size = rare_size, prob = rare_prob, lower.tail = FALSE),
    ppois(5, lambda = poisson_lambda, lower.tail = FALSE)
  )
)

knitr::kable(
  rare_summary,
  digits = 5,
  booktabs = TRUE,
  caption = "Rare-event summary comparison"
)
```

Table 5: Rare-event summary comparison

source	mean	variance	probability_at_least_6
simulated binomial	3.00654	3.00483	0.08437
exact binomial	3.00000	2.99100	0.08361
Poisson comparison	3.00000	3.00000	0.08392

The exact binomial and Poisson upper-tail probabilities differ by only 0.00030. The approximation works well because there are many trials and a small success probability.

4 Checking Poisson equidispersion

> **Task:** Repeat 1,000 times: simulate 200 counts from Poisson(4) and record their sample mean and variance. Plot variance against mean. Then repeat after drawing each count's rate independently from $\{1, 7\}$ with equal probability. Compare the plots and explain why the heterogeneous counts are overdispersed relative to a single Poisson model.

For the homogeneous model, every count has the same rate:

$$X_i \sim \text{Poisson}(4).$$

Its population mean and variance are both 4. We therefore expect each dataset's sample mean and variance to fluctuate around the diagonal point $(4, 4)$.

For the heterogeneous model, each observation first receives rate 1 or rate 7 with equal probability. The average rate remains

$$\frac{1+7}{2} = 4.$$

However, the rates themselves vary. Their variance is

$$\frac{(1-4)^2 + (7-4)^2}{2} = 9.$$

That between-rate variation adds to the average within-rate Poisson variance of 4, giving an overall count variance of

$$4 + 9 = 13.$$

Thus, both models have mean 4, but only the homogeneous model has variance 4.

```r
set.seed(504)

count_reps <- 1000
counts_per_dataset <- 200

simulate_count_summaries <- function(repetitions, draw_counts) {
  results <- replicate(repetitions, {
    counts <- draw_counts()
    c(
      sample_mean = mean(counts),
      sample_variance = var(counts)
    )
  })

  as.data.frame(t(results))
}

homogeneous <- simulate_count_summaries(
  count_reps,
  function() rpois(counts_per_dataset, lambda = 4)
) |>
  mutate(model = "single rate: lambda = 4")

heterogeneous <- simulate_count_summaries(
  count_reps,
  function() {
    rates <- sample(
      c(1, 7),
      size = counts_per_dataset,
      replace = TRUE
    )
    rpois(counts_per_dataset, lambda = rates)
  }
) |>
```

```
  mutate(model = "mixed rates: 1 or 7")

poisson_checks <- bind_rows(homogeneous, heterogeneous)
```

```
poisson_summary <- poisson_checks |>
  group_by(model) |>
  summarise(
    average_sample_mean = mean(sample_mean),
    average_sample_variance = mean(sample_variance),
    sd_of_sample_means = sd(sample_mean),
    .groups = "drop"
  )

poisson_summary_display <- poisson_summary |>
  transmute(
    Model = model,
    `Average mean` = average_sample_mean,
    `Average variance` = average_sample_variance,
    `SD of means` = sd_of_sample_means
  )

knitr::kable(
  poisson_summary_display,
  digits = 3,
  booktabs = TRUE,
  caption = "Repeated-simulation summaries"
)
```

Table 6: Repeated-simulation summaries

Model	Average mean	Average variance	SD of means
mixed rates: 1 or 7	4.012	13.038	0.257
single rate: lambda = 4	3.998	3.991	0.138

```
poisson_checks |>
  ggplot(aes(sample_mean, sample_variance)) +
  geom_abline(
    slope = 1,
    intercept = 0,
    color = "#d95f02",
    linewidth = 0.6,
    linetype = "dashed"
  ) +
  geom_point(color = "#3564d4", alpha = 0.25, size = 0.9) +
  facet_wrap(~ model) +
  scale_x_continuous(
    limits = c(0, 18),
    breaks = seq(0, 18, by = 3),
```

```
    expand = expansion(mult = 0.02)
  ) +
  scale_y_continuous(
    limits = c(0, 18),
    breaks = seq(0, 18, by = 3),
    expand = expansion(mult = 0.02)
  ) +
  coord_fixed(ratio = 1) +
  labs(
    x = "sample mean",
    y = "sample variance"
  )
```

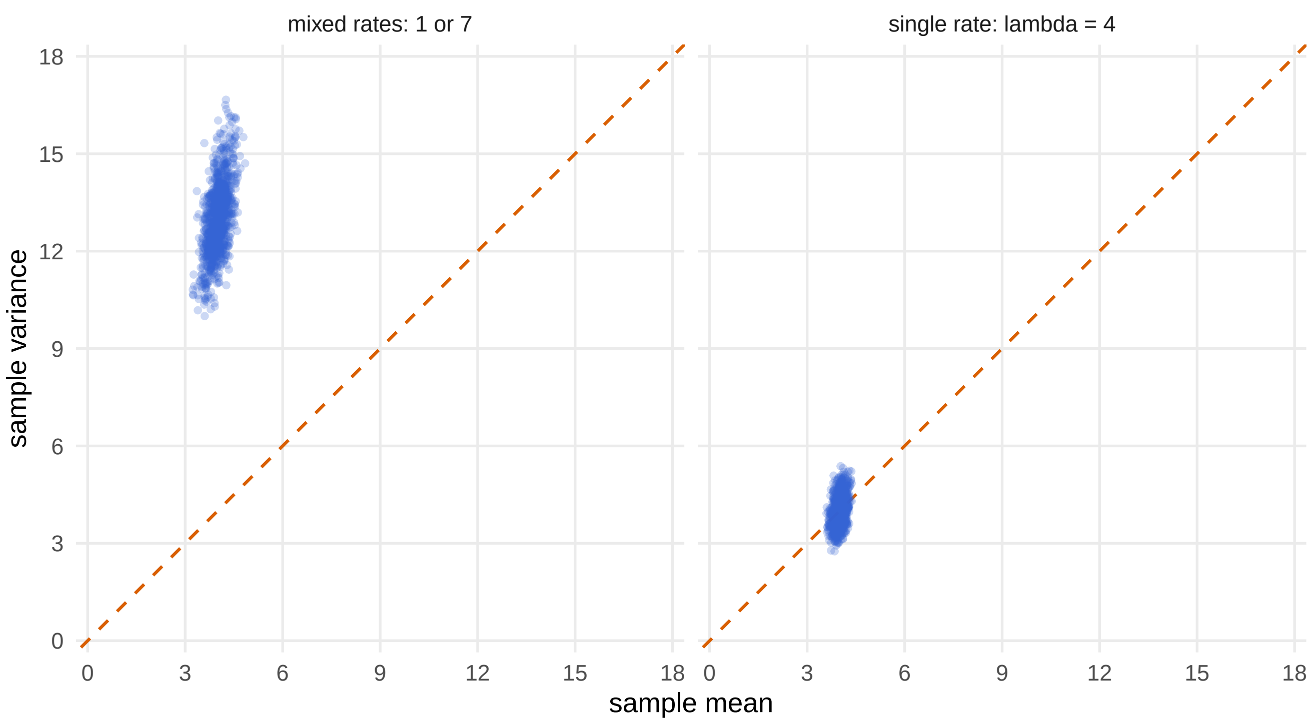

Figure 4: Sample variance versus sample mean for homogeneous and heterogeneous Poisson counts.

Both axes now run from 0 to 18 with a 1:1 aspect ratio, so equal numerical changes occupy equal visual distances. The nearly vertical clouds show that the sample means vary within a narrow range. The homogeneous simulations cluster near mean 4 and variance 4, close to the dashed equidispersion line. The heterogeneous simulations retain mean 4 but cluster near variance 13. A single Poisson model with rate 4 cannot reproduce that extra variation, so the heterogeneous counts are overdispersed relative to it.

5 Main conclusions

1. `dbinom()` evaluates a binomial point probability, while `pbinom()` accumulates probabilities through a cutoff.

2. A geometric simulation using `rgeom() + 1` agrees with the first-success PMF, mean, and variance.
3. A Poisson PMF closely approximates a binomial PMF when the number of trials is large, the success probability is small, and $N\theta$ is held fixed.
4. Poisson equidispersion can fail when observations have different rates, even when their average rate is unchanged.